1 在谐振子与谐振子热库模型中利用Influence Functional推导主方程

谐振子自由度x，谐振子热库自由度q_i. Assume the bath has a temperature β.

$$S[x,q] = S[x] + S[q_i] + S_I[x,q_i] \tag{1}$$

$$S[x] = \int_0^t ds \frac{1}{2}M\dot{x}^2 - \frac{1}{2}M\Omega^2 x^2, \tag{2}$$

$$S[q_i] = \int_0^t ds \frac{1}{2}m_i\dot{q}_i^2 - \frac{1}{2}m_i\omega_i^2 q_i^2, \tag{3}$$

$$S_I[x,q_i] = -\int ds \sum_i c_i x * q_i. \tag{4}$$

1.1 影响泛函的具体形式

Our system and bath interaction action is $S[x,q]$.

$$\rho_r(x_f, x_f', t) = \int dx_i \int dx_i' J_r(x_f, x_f'|x_i, x_i, t)\rho_r(x_i, x_i', 0)$$
$$J_r(x_f, x_f'|x_i, x_i, t) = \int \mathcal{D}x \int \mathcal{D}x' e^{\mathrm{i}(S[x(s)]-S[x'(s)]+\delta A[x,x'])}$$

And I need to figure out the form of δA.

1.2 热库自由度q_i的关联函数

$$\langle q_n q_m \rangle_\beta \tag{5}$$

Define collected coordinate $Q(s) = \sum_n c_n q_n(s)$, because it interacts with system coordinate x with the form $x(s)\sum_n c_n q_n(s) = x(s)Q(s)$.
In the case of ocassilator bath, $\langle Q \rangle_\beta = 0$. No shift, and disconnect correlation function equal to connect correlation function.

$$\begin{aligned}\langle Q(s)Q(s')\rangle_\beta = \int_0^\infty d\omega J(\omega)\left[\coth(\frac{\beta\hbar\omega}{2})\cos(\omega(s-s')) - \mathrm{i}\sin\omega(s-s')\right] \\ = \nu(s-s') + \mathrm{i}\eta(s-s') \\ J(\omega) = \sum_n \delta(\omega-\omega_n)\frac{c_n^2}{2m\omega_n}\end{aligned} \tag{6}$$

The real part of the correlation function is fluctuation kernel, and the imaginary part is dissipation kernel $\eta(s-s') = \mathrm{Im}\,\langle Q(s)Q(s')\rangle_\beta$.

This structure origins from the average on reservoir DOF: $\langle \exp\left\{\left(\mathrm{i}\int x(s)Q(s) - x'(s)Q'(s)\right)\right\}\rangle_{Q,Q'} = \exp\left\{(\mathrm{i}\delta A[x,x'])\right\}$

$$\begin{aligned}\delta A[x,x'] = -\int_0^t ds[x(s)\langle Q(s)\rangle - x'(s)\langle Q'(s)\rangle] + \frac{\mathrm{i}}{2}\int ds \int ds' x(s)\langle \mathcal{T}Q(s)Q(s')\rangle x(s') + x'(s)\langle \bar{\mathcal{T}}Q(s)Q(s')\rangle x'(s') \\ -x(s)\langle Q(s')Q(s)\rangle x'(s') - x(s')\langle Q(s)Q(s')\rangle x'(s) + \cdots\end{aligned} \tag{7}$$

$$
\begin{aligned}
\delta A[x,x'] &= \mathrm{i}\int_0^t ds \int_0^s ds'(x(s)-x'(s))\nu(s-s')(x(s')-x'(s')) + \mathrm{i}(x(s)-x'(s))\eta(s-s')(x(s')+x'(s')) \\
&= \mathrm{i}\int_0^t ds_1 \int_0^{s_1} ds_2 \Delta(s_1)\nu(s_1-s_2)\Delta(s_2) - 2\int_0^t ds_1 \int_0^{s_1} ds_2 \Delta(s_1)\eta(s_1-s_2)\Sigma(s_2) \\
&\qquad \text{where} \quad \Delta(s) = x(s)-x'(s), \Sigma(s) = \frac{x(s)+x'(s)}{2}.
\end{aligned} \tag{8}
$$

Given a symbol $W_+(s-s') = \langle Q(s)Q(s')\rangle_\beta$, and $W_-(s-s') = \langle Q(s')Q(s)\rangle_\beta$, which satisfy $W_+(s-s') = W_-(s'-s)$.

In order to get the formula，I split the integral region into two part to make the time-order explicitly.

1.3 利用影响泛函求主方程

Main idea: The master equation is obtained by evaluating $J_r(t+\Delta t)$, subtracting $J_r(t)$, and expanding the difference to first order in Δt, retaining terms up to $\mathcal{O}(\Delta t)$ while discarding higher-order contributions.

1.3.1 插入t时刻中间坐标x_m, x'_m

Spit the trajectory into two part $\bar{x}, \tilde{x}$. Integrate $\tilde{x}$ with line approximation:$\tilde{x}(s) = x_m + (x_f - x_m)\frac{s}{\Delta t}$.

$$
\int_{0,x_i}^{t+\Delta t,x_f} \mathcal{D}x = \int_{-\infty}^{+\infty} dx_m \int_{0,x_i}^{t,x_m} \mathcal{D}\bar{x} \int_{t,x_m}^{t+\Delta t,x_f} \mathcal{D}\tilde{x} \tag{9}
$$

$$
\begin{aligned}
J_r(t+\Delta t; x_f, x'_f) = N(t)\int_0^\infty dx_m \int_0^\infty dx'_m \exp\left\{[\mathrm{i}\frac{M}{2\Delta t}((x_f-x_m)^2-(x'_f-x'_m)^2)]\right\} \cdot \left[1-\mathrm{i}\Delta t(V(x_f)-V(x'_f))\right] \\
\cdot \int_i^m \mathcal{D}\bar{x}\mathcal{D}\bar{x}' \exp\left\{[\mathrm{i}(S[\bar{x}]-S[\bar{x}'])]\right\} \exp\left\{[\mathrm{i}\delta A[\bar{x},\bar{x}',\tilde{x},\tilde{x}']]\right\}
\end{aligned} \tag{10}
$$

1.3.2 $\delta A[\bar{x},\bar{x}',\tilde{x},\tilde{x}']$形式

$$
\begin{aligned}
\delta A[\bar{x},\bar{x}',\tilde{x},\tilde{x}'] &= -2\int_0^{t+\Delta t} ds_1 \int_0^{s_1} ds_2 \Delta(s_1)\eta(s_1-s_2)\Sigma(s_2) + \mathrm{i}\int_0^{t+\Delta t} ds_1 \int_0^{s_1} ds_2 \Delta(s_1)\nu(s_1-s_2)\Delta(s_2) \\
&= -2\int_0^t ds_1 \int_0^{s_1} ds_2 \bar{\Delta}(s_1)\eta(s_1-s_2)\bar{\Sigma}(s_2) - 2\int_t^{t+\Delta t} ds_1 \int_0^t ds_2 \tilde{\Delta}(s_1)\eta(s_1-s_2)\bar{\Sigma}(s_2) \\
&\quad -2\int_t^{t+\Delta t} ds_1 \int_t^{s_1} ds_2 \tilde{\Delta}(s_1)\eta(s_1-s_2)\tilde{\Sigma}(s_2) + \mathrm{i}\int_0^t ds_1 \int_0^{s_1} ds_2 \bar{\Delta}(s_1)\nu(s_1-s_2)\bar{\Delta}(s_2) \\
&\quad +\mathrm{i}\int_t^{t+\Delta t} ds_1 \int_0^t ds_2 \tilde{\Delta}(s_1)\nu(s_1-s_2)\bar{\Delta}(s_2) + \mathrm{i}\int_t^{t+\Delta t} ds_1 \int_t^{s_1} ds_2 \tilde{\Delta}(s_1)\nu(s_1-s_2)\tilde{\Delta}(s_2)
\end{aligned} \tag{11}
$$

There are three types of terms: those of order $\mathcal{O}(\Delta t^0)$, those of order $\mathcal{O}(\Delta t^1)$, and those of order $\mathcal{O}(\Delta t^2)$, the latter of which will be neglected because it is higher order of Δt. Backing to the Reduced propagator

J_r:

$$J_r(t+\Delta t;x_f,x_f')=N(t)\int_0^\infty dx_m\int_0^\infty dx_m'\exp\left\{[\mathrm{i}\frac{M}{2\Delta t}((x_f-x_m)^2-(x_f'-x_m')^2)]\right\}\cdot\left[1-\mathrm{i}\Delta tV(x_f)-\mathrm{i}\Delta tV(x_f')\right]$$
$$\cdot\int_i^m\mathcal{D}\bar{x}\mathcal{D}\bar{x}'\exp\left\{[\mathrm{i}\left(S[\bar{x}]-S[\bar{x}']-2\int_0^t ds_1\int_0^{s_1}ds_2\bar{\Delta}(s_1)\eta(s_1-s_2)\bar{\Sigma}(s_2)+\mathrm{i}\int_0^t ds_1\int_0^{s_1}ds_2\bar{\Delta}(s_1)\nu(s_1-s_2)\bar{\Delta}(s_2)\right)]\right\}$$
$$\cdot\exp\left\{\mathrm{i}\Delta t\left[-2\int_0^t ds\bar{\Sigma}(s)J_\Sigma(s)+\mathrm{i}\int_0^t ds\bar{\Delta}(s)J_\Delta(s)\right]\right\}$$
$$\text{where}\quad J_\Sigma(s)=\frac{1}{\Delta t}\int_t^{t+\Delta t}ds_1\tilde{\Delta}(s_1)\eta(s_1-s),J_\Delta(s)=\frac{1}{\Delta t}\int_t^{t+\Delta t}ds_1\tilde{\Delta}(s_1)\nu(s_1-s).\tag{12}$$

The approximate expressions of the source are $J_\Sigma(s)\approx\frac{1}{2}(\Delta_f+\Delta_m)\eta(t-s)$,$J_\Delta(s)\approx\frac{1}{2}(\Delta_f+\Delta_m)\nu(t-s)$. The last could be seen as reduced propagator$J_r(t;x_m,x_m';J_\Delta,J_\Sigma)$ with source. I call the $(\Delta t)^0$ part of action $A_0[x,x']$.

$$A_0[\bar{\Sigma},\bar{\Delta}]=\int_0^t dsM(\dot{\bar{\Delta}}\dot{\bar{\Sigma}}-\Omega^2\bar{\Delta}\bar{\Sigma})$$
$$-2\int_0^t ds_1\int_0^{s_1}ds_2\bar{\Delta}(s_1)\eta(s_1-s_2)\bar{\Sigma}(s_2)$$
$$+\mathrm{i}\int_0^t ds_1\int_0^{s_1}ds_2\bar{\Delta}(s_1)\nu(s_1-s_2)\bar{\Delta}(s_2).$$

1.3.3 利用鞍点近似在Δt阶完成$\bar{x}$积分

$$\Sigma(s)=\chi_1(s)+\Sigma_{\mathrm{cl}}(s),\qquad\Delta(s)=\chi_2(s)+\Delta_{\mathrm{cl}}(s).$$

the classical part satisfy equation $\frac{\delta\,\mathrm{Re}\,A_0}{\delta(*)}=0$,

$$\ddot{\Sigma}_{\mathrm{cl}}(s)+\Omega^2\Sigma_{\mathrm{cl}}(s)+2\int_0^s ds'\,\eta(s-s')\Sigma_{\mathrm{cl}}(s')=0,\tag{13}$$
$$\ddot{\Delta}_{\mathrm{cl}}(s)+\Omega^2\Delta_{\mathrm{cl}}(s)+2\int_s^t ds'\,\eta(s'-s)\Delta_{\mathrm{cl}}(s')=0,\tag{14}$$

carefully handle the exponential part in J_r. Expanding J_r around saddle point Σ_{cl},Δ_{cl}.

$$A_0[\Sigma_{cl}+\chi_1,\Delta_{cl}+\chi_2]=A_0[\Sigma_{cl},\Delta_{cl}]+\mathrm{i}\int_0^t ds\int_0^s ds'2\chi_2(s)\nu(s-s')\Delta_{cl}(s')$$
$$+\int_0^t dsM\left[\dot{\chi}_1\dot{\Delta}_{cl}+\dot{\chi}_2\dot{\Sigma}_{cl}-\Omega^2\chi_1\Delta_{cl}-\Omega^2\chi_2\right]-2\int_0^t ds\int_0^s ds'\left[\chi_2(s)\eta(s-s')\Sigma_{cl}(s')+\Delta_{cl}(s)\eta(s-s')\chi_1(s')\right]$$
$$+\int_0^t M[\dot{\chi}_1\dot{\chi}_2-\Omega^2\chi_1\chi_2]-2\int_0^t ds\int_0^s ds'\chi_2(s)\eta(s-s')\chi_1(s')+\mathrm{i}\int_0^t ds\int_0^s ds'\chi_2(s)\nu(s-s')\chi_2(s')$$

the second line vanish because the saddle points condition.

$$\delta A[\Sigma,\Delta;\tilde{\Sigma},\tilde{\Delta}]=\Delta t\left[-2\int_0^t dsJ_\Sigma(s)(\chi_1(s)+\Sigma_{cl})+\mathrm{i}\int_0^t dsJ_\Delta(s)(\chi_2(s)+\Delta_{cl}(s))\right]$$

$$J_r[m;i;J_{\Sigma,\Delta}]=\exp\left\{\mathrm{i}(A_0[\Sigma_{cl},\Delta_{cl}]+\delta A[\Sigma_{cl},\Delta_{cl}])\right\}\underbrace{\int_0^0\mathcal{D}\chi_1\int_0^0\mathcal{D}\chi_2\text{quadratic form of}\chi}_{J_r[0,0,t|0,0;\text{source}]}\tag{15}$$

$$J_r[0,0,t|0,0;\text{source}] = \int_0^0 \mathcal{D}\chi_1 \int_0^0 \mathcal{D}\chi_2 \exp\left\{ \mathrm{i}\left[\int_0^t ds_1 \int_0^t ds_2 \frac{1}{2}\chi_i(s_1)\mathcal{O}_{ij}(s_1,s_2)\chi_i(s_2) + \int_0^t ds \chi_i B_i \right]\right\}$$

$$O_{11}(s_1,s_2) = 0, \qquad O_{22}(s_1,s_2) = 2i\,\nu(s_1,s_2),$$

$$O_{12}(s_1,s_2) = \left[\frac{d^2}{ds_1^2} + \Omega^2\right]\delta(s_1-s_2) + 2\theta(s_2-s_1)\,\eta(s_2-s_1),$$

In order to figure out the O_{12}, we split the -2 in the third line into $-1-1$. And exchange the variables s_1, s_2 in the second -1.

$$O_{21}(s_1,s_2) = \left[\frac{d^2}{ds_1^2} + \Omega^2\right]\delta(s_1-s_2) + 2\theta(s_1-s_2)\,\eta(s_1-s_2).$$

$$B_1(s) = -2\Delta t J_\Sigma(s) \qquad B_2(s) = \mathrm{i}\Delta t J_\Delta(s) + \mathrm{i}\underbrace{\int_0^s ds' 2\nu(s-s')\Delta_{cl}(s')}_{\tilde{J}_\Delta}$$

Do the Gaussian integral with formula $\int \mathcal{D}\varphi \exp\left\{\mathrm{i}\int dx \int dy \frac{1}{2}\varphi(x)K(x-y)\varphi(y) + \mathrm{i}\int_x J(x)\varphi(x)\right\} \propto \exp\left\{-\mathrm{i}\frac{1}{2}\int_x\int_y J(x)K^{-1}(\right.$

$$J_r[0,0,t|0,0;\boldsymbol{B}] = \mathcal{N}\exp\left\{-\mathrm{i}\frac{1}{2}\int_0^t ds \int_0^t ds' B_i(s)G_{ij}(s,s')B_j(s')\right\} \tag{16}$$

where the kernel operator matrix G_{ij} is the reversal of $\mathcal{O}$. And satisfy boundary condition with χ: $G(0,s) = 0 = G(s,t) = G(s,0) = G(t,s)$. And satisfy Green equation:

$$\begin{aligned}
\int_0^t ds_2\, O_{12}(s_1,s_2)G_{21}(s_2,s_3) &= \delta(s_1-s_3)\\
\left(\partial_{s_1}^2 + \Omega^2\right)G_{21}(s_1,s_3) + 2\int_0^{s_1} ds_2\,\eta(s_1-s_2)G_{21}(s_2,s_3) &= \delta(s_1-s_3)\\
\int_0^t ds_2\, O_{21}(s_1,s_2)G_{12}(s_2,s_3) &= \delta(s_1-s_3)\\
\left(\partial_{s_1}^2 + \Omega^2\right)G_{12}(s_1,s_3) + 2\int_{s_1}^t ds_2\,\eta(s_2-s_1)G_{12}(s_2,s_3) &= \delta(s_1-s_3)
\end{aligned} \tag{17}$$

I keep only terms up to $\mathcal{O}(\Delta t)$. The term $B_1 G_{11} B_1$ is of order $(\Delta t)^2$ and is therefore discarded. And the result of $J_r[0,0,t|0,0;\boldsymbol{B}]$ is :

$$J_r[0,0,t|0,0;\boldsymbol{B}] = \mathcal{N}\exp\left\{-\frac{1}{2}\Delta t\int_0^t ds_1 \int_0^t ds_2 J_\Sigma(s_1)(G_{12}(s_1.s_2) + G_{21}(s_2,s_1))\tilde{J}_\Delta(s_2)\right\} \tag{18}$$

$$J_r[m;i;J_{\Sigma,\Delta}] = \underbrace{\mathcal{N}\exp\left\{\mathrm{i}(A_0[\Sigma_{cl},\Delta_{cl}])\right\}}_{\approx J_r[x_m,x'_m,t;x_i,x'_i]}$$

$$\cdot\exp\left\{\mathrm{i}\Delta t\left[-2\int_0^t ds J_\Sigma(s)\Sigma_{cl}(s) + \mathrm{i}\int_0^t ds J_\Delta(s)\Delta_{cl}(s)\right] - \frac{1}{2}\Delta t\int_0^t ds_1\int_0^t ds_2 J_\Sigma(s_1)(G_{12}(s_1.s_2) + G_{21}(s_2,s_1))\tilde{J}_\Delta(s_2)\right\}$$

The x_m, x'_m dependence is hidden in Σ_{cl}, Δ_{cl}. The f dependence is hidden in source $J_{\Sigma/\Delta}$.

1.3.4 处理Σ_{cl},Δ_{cl}与G

I can't solve (13)(14) analytically, because they are integro-differential equation with memory kernel. I just make the boundary condition explicitly. Write the solutions of (13) as $u_1(s)$ and $u_2(s)$ with conditions

$u_1(s=0)=1=u_2(s=t)$; $\quad u_1(s=t)=0=u_2(s=0)$. In this way , the saddle can be expressed as $\Sigma_{cl}(s)=\Sigma_i u_1(s)+\Sigma_f u_2(s)$. And define $v_1(s)=u_2(t-s), v_2(s)=u_1(t-s)$, which is the solution of (14). $\Delta_{cl}(s)=\Delta_i v_1(s)+\Delta_m v_2(s)$.

By the way. Without the memory kernel, the saddles are exact, $\Delta_{cl}(s)=\Delta_i\frac{\sin\Omega(t-s)}{\sin\Omega t}+\Delta_f\frac{\sin\Omega s}{\sin\Omega t}$.

The green equation (17) is similar to (14). We can construct solution： $G_{12}(s_1,s_2)=[u_1(s_1)u_2(s_2)\theta(s_1-s_2)-u_2(s_1)u_1(s_2)\theta(s_2-s_1)](\dot{u}_1u_2-u_1\dot{u}_2)^{-1}(s_2)$.

1.3.5 整理$J_r(t+\Delta t)$

$$J_r(t+\Delta t;x_f,x_f')=N(t)\int_{-\infty}^{\infty}dx_m\int_{-\infty}^{\infty}dx_m'\exp\left\{[\mathrm{i}\frac{M}{2\Delta t}(\beta_x^2-\beta_x'^2)]\right\}\cdot\left[1-\mathrm{i}\Delta tV(x_f)-\mathrm{i}\Delta tV(x_f')\right]\cdot J_r(t;x_m,x_m')$$
$$\cdot\exp\left\{\mathrm{i}\Delta t\left[-2\int_0^t dsJ_\Sigma(s)\Sigma_{cl}(s)+\mathrm{i}\int_0^t dsJ_\Delta(s)\Delta_{cl}(s)\right]-\frac{1}{2}\Delta t\int_0^t ds_1\int_0^t ds_2J_\Sigma(s_1)(G_{12}(s_1.s_2)+G_{21}(s_2,s_1))\tilde{J}_\Delta(s_2)\right\}$$

where $\beta_x=x_f-x_m, x_m=x_f-\beta_x$, thus $J_r(t;x_m,x_m')=J_r(t;x_f,x_f')+\frac{\beta_x^2}{2}\partial_x^2J_r+\frac{\beta_{x'}^2}{2}\partial_{x_f'}^2J_r$.

$$\begin{aligned}\int d\beta_xe^{\mathrm{i}\frac{M}{2\Delta t}\beta_x^2}&=\sqrt{\frac{2\pi\mathrm{i}\Delta t}{M}}=N_1,\quad\int d\beta_xe^{\mathrm{i}\frac{M}{2\Delta t}\beta_x^2}\beta_x=0,\quad\int d\beta_x\beta_x^2e^{\mathrm{i}\frac{M}{2\Delta t}\beta_x^2}=\sqrt{\frac{2\pi\mathrm{i}\Delta t}{M}}\frac{\mathrm{i}\Delta t}{M}.\\ \int d\beta_xe^{-\mathrm{i}\frac{M}{2\Delta t}\beta_x^2}&=\sqrt{\frac{-2\pi\mathrm{i}\Delta t}{M}}=N_2,\quad\int d\beta_xe^{-\mathrm{i}\frac{M}{2\Delta t}\beta_x^2}\beta_x=0,\quad\int d\beta_x\beta_x^2e^{-\mathrm{i}\frac{M}{2\Delta t}\beta_x^2}=\sqrt{\frac{-2\pi\mathrm{i}\Delta t}{M}}\frac{-\mathrm{i}\Delta t}{M}\end{aligned}\tag{19}$$

$$N\int d\beta_x\int d\beta_{x'}\exp\left\{[\mathrm{i}\frac{M}{2\Delta t}(\beta_x^2-\beta_x'^2)]\right\}(\frac{\beta_x^2}{2}\partial_x^2J_r+\frac{\beta_{x'}^2}{2}\partial_{x_f'}^2J_r)=\frac{\mathrm{i}}{2}(\partial_x^2J_r)\frac{\Delta t}{M}-\frac{i}{2}(\partial_{x'}^2J_r)\frac{\Delta t}{M}$$

(19) explained that in the $x_m(\beta_x)$ integral, x_m equal to x_f. The width of (19) is $\sqrt{\Delta t}$, and β^2 term gives a contribution $\mathcal{O}(\Delta t)$. And the normalization coefficients satisfy $N*N_1*N_2=1$.

$$\begin{aligned}-2\int_0^t dsJ_\Sigma(s)\Sigma_{cl}(s)&=-2\int ds(\Delta_f\eta(t-s))(\Sigma_iu_1(s)+\Sigma_fu_2(s))\\&=-2\left(\Delta_f\Sigma_i\int_0^t ds\eta(t-s)u_1(s)+\Delta_f\Sigma_f\int_0^t ds\eta(t-s)u_2(s)\right)=-(\Delta_f\Sigma_id_1+\Delta_f\Sigma_fd_2)\end{aligned}$$

where $d_i=2\int_0^t ds\eta(t-s)u_i(s)$ and $e_i=\int_0^t ds\nu(t-s)v_{j\neq i}(s)=\int_0^t ds\nu(s)u_i(s)$.

$$\begin{aligned}\int_0^t dsJ_\Delta(s)\Delta_{cl}(s)&=\int ds\Delta_f\nu(t-s)(\Delta_iv_1(s)+\Delta_fv_2(s))\\&=\Delta_i\Delta_f\int_0^t ds\nu(s)u_2(s)+\Delta_f^2\int ds\nu(s)u_1(s)=\Delta_i\Delta_fe_2+\Delta_f^2e_1\end{aligned}$$

$\tilde{J}_\Delta(s)=\int_0^s ds'2\nu(s-s')\Delta_{cl}(s')=2\int_0^s ds'\nu(s-s')(\Delta_iv_1(s')+\Delta_fv_2(s'))$.

$$\begin{aligned}-\frac{1}{2}\int_0^t ds_1\int_0^t ds_2J_\Sigma(s_1)(G_{12}(s_1.s_2)+G_{21}(s_2,s_1))\tilde{J}_\Delta(s_2)&=-\Delta_i\Delta_fc_1-\Delta_f^2c_2\\=-\Delta_i\Delta_f\int_0^t ds_1\int_0^t ds_2\int_0^{s_2}ds_3\eta(t-s_1)(G_{12}(s_1,s_2)+G_{21}(s_2,s_1))\nu(s_2-s_3)v_1(s_3)&\\-\Delta_f^2\int_0^t ds_1\int_0^t ds_2\int_0^{s_2}ds_3\eta(t-s_1)(G_{12}(s_1,s_2)+G_{21}(s_2,s_1))\nu(s_2-s_3)v_2(s_3)&\end{aligned}$$

$c_i(t) = \int_0^t ds_1 \int_0^t ds_2 \int_0^{s_2} ds_3 \eta(t-s_1)(G_{12}(s_1,s_2)+G_{21}(s_2,s_1))\nu(s_2-s_3)v_i(s_3)$; c_i comes from fluctuation integration of χ, and it is order $\mathcal{O}(\lambda^4)$.
In the order $\mathcal{O}(\Delta t)$:

$$J_r(t+\Delta t; x_f, x_f') = N(t)\int_{-\infty}^{\infty} dx_m \int_{-\infty}^{\infty} dx_m' \exp\left\{[\mathrm{i}\frac{M}{2\Delta t}(\beta_x^2-\beta_x'^2)]\right\}\cdot\left[1-\mathrm{i}\Delta t V(x_f)-\mathrm{i}\Delta t V(x_f')\right]$$
$$\cdot(J_r(t;x_f,x_f')+\frac{\beta_x^2}{2}\partial_x^2 J_r+\frac{\beta_{x'}^2}{2}\partial_{x_f'}^2 J_r)\cdot\left(1-\mathrm{i}\Delta t(\Delta_f\Sigma_i d_1+\Delta_f\Sigma_f d_2)-\Delta_i\Delta_f(e_2+c_1)\Delta t-\Delta_f^2(e_1+c_2)\Delta t\right)$$

$$\partial_t J_r(t) = -\mathrm{i}\left[-\frac{1}{2}(\partial_x^2-\partial_{x'}^2+\Omega^2(x_f^2-x_f'^2))\right]J_r + [\text{Terms with}\Delta_i,\Sigma_f,\Delta_f]J_r$$

$$\begin{aligned}J_r(t+\Delta t;x,x') =& J_r(t;x,x')+\frac{\mathrm{i}}{2}(\partial_x^2 J_r)\frac{\Delta t}{M}-\frac{i}{2}(\partial_{x'}^2 J_r)\frac{\Delta t}{M}\\ &+J_r(t,;x,x')\Delta t\left(-\mathrm{i}V(x)+\mathrm{i}V(x')-\mathbf{i}\Delta_f\Sigma_i d_1-\mathrm{i}\Delta_f\Sigma_f d_2-\Delta_i\Delta_f(e_2+c_1)-\Delta_f^2(e_1+c_2)\right)\end{aligned}$$

$$\mathrm{i}\partial_t J_r(t;x,x') = \left\{-\frac{1}{2M}(\partial_x^2-\partial_{x'}^2)+V(x)-V(x')+\Delta_f\Sigma_i d_1+\Delta_f\Sigma_f d_2-\mathrm{i}\Delta_i\Delta_f(e_2+c_1)-\mathrm{i}\Delta_f^2(e_1+c_2)\right\}J_r(t;x,x') \tag{20}$$

1.3.6 处理初始条件相关变量Σ_i, Δ_i

In order to determine the expressions for $\Sigma_i J_r(t)$ and $\Delta_i J_r(t)$, we examine the explicit form of J_r. $J_r(t,f;i,0) = \mathcal{N}e^{\mathrm{i}A_0[\Sigma_{cl},\Delta_{cl}]}$,$A_0[\Sigma_{cl},\Delta_{cl}] = \int_0^t ds M(\dot{\Sigma}_{cl}\dot{\Delta}_{cl}-\Omega^2\Delta_{cl}\Sigma_{cl}) - 2\int_0^t ds_1\int_0^{s_1} ds_2 \Delta_{cl}(s_1)\eta(s_1-s_2)\Sigma_{cl}(s_2)+\frac{1}{2}\mathrm{i}\int_0^t ds_1\int_0^t ds_2\Delta_{cl}(s_1)\nu(s_1-s_2)\Delta_{cl}(s_2)$. Do partial integral: $A_0[\Sigma_{cl},\Delta_{cl}] = M(\Delta_f\dot{\Sigma}_f-\Delta_i\dot{\Sigma}_i)+\mathrm{i}\,\mathrm{Im}\,A_0[\Delta_{cl}]$. We can get an expression of J_r which depends on boundary conditions explicitly.

$$\begin{aligned}J_r = \mathcal{N}\exp\{\mathrm{i}M[\Delta_f\Sigma_i\dot{u}_1(t)-\Delta_i\Sigma_i\dot{u}_1(0)-\Delta_i\Sigma_f\dot{u}_2(0)+\Delta_f\Sigma_f\dot{u}_2(t)]\}\\ \cdot\exp\left\{-\frac{1}{2}(\Delta_i^2 a_{11}+\Delta_i\Delta_f(a_{12}+a_{21})+\Delta_f^2 a_{22})\right\}\end{aligned}$$

where $a_{ij} = \int_0^t ds_1\int_0^t ds_2 v_i(s_1)\nu(s_1-s_2)v_2(s_2)$.

$$\partial_{\Sigma_f}J_r = [-\mathrm{i}\Delta_i\dot{u}_2(0)+\mathrm{i}\Delta_f\dot{u}_2(t)]J_r$$
$$\Delta_i J_r = (\frac{\dot{u}_2(t)}{\dot{u}_2(0)}\Delta_f+\mathrm{i}\frac{1}{\dot{u}_2(0)}\partial_{\Sigma_f})J_r$$

$$\Sigma_i J_r(t,0) = \left[-\frac{\dot{u}_2(t)}{\dot{u}_1(t)}\Sigma_f - i\left(\frac{4a_{22}(t)}{\dot{u}_1(t)}+\frac{2a_{21}(t)}{\dot{u}_1(t)}\frac{\dot{u}_2(t)}{\dot{u}_2(0)}\right)\Delta_f - \frac{i}{\dot{u}_1(t)}\frac{\partial}{\partial\Delta_f}+\frac{2a_{21}(t)}{\dot{u}_1(t)\dot{u}_2(0)}\frac{\partial}{\partial\Sigma_f}\right]J_r(t,0),$$

1.3.7 将$\Delta_f, \Sigma_f, \partial_\Delta, \partial_\Sigma$替换为$x, x'$

Definition：$\Sigma_f = \frac{1}{2}(x+x'), \Delta_f = x-x'$; inverse transform $x = \Sigma_f+\frac{1}{2}\Delta_f, x' = \Sigma_f-\frac{1}{2}\Delta_f$; derivative relation $\partial_\Sigma = \partial_x+\partial_{x'}, \partial_\Delta = \frac{1}{2}(\partial_x-\partial_{x'})$. We get the replacement rule the:

$$\Delta_f^2 J_r(t) = (x-x')^2 J_r(t)$$
$$\Delta_f\Sigma_f J_r(t) = \frac{1}{2}(x^2-x'^2)J_r(t)$$
$$\Delta_f\partial_{\Sigma_f}J_r = (x-x')(\partial_x+\partial_{x'})J_r$$
$$\Delta_f\partial_{\Delta_f}J_r = \frac{1}{2}(x-x')(\partial_x-\partial_{x'})J_r.$$

$$\Delta_f\Delta_i J_r = \Delta_f(\frac{\dot{u}_2(t)}{\dot{u}_2(0)}\Delta_f + \mathrm{i}\frac{1}{\dot{u}_2(0)}\partial_{\Sigma_f})J_r$$

$$\Delta_f\Sigma_i J_r = \Delta_f\left[-\frac{\dot{u}_2(t)}{\dot{u}_1(t)}\Sigma_f - i\left(\frac{4a_{22}(t)}{\dot{u}_1(t)} + \frac{2a_{21}(t)}{\dot{u}_1(t)}\frac{\dot{u}_2(t)}{\dot{u}_2(0)}\right)\Delta_f - \frac{i}{\dot{u}_1(t)}\frac{\partial}{\partial\Delta_f} + \frac{2a_{21}(t)}{\dot{u}_1(t)\dot{u}_2(0)}\frac{\partial}{\partial\Sigma_f}\right]J_r(t,0),$$

In (20), collect the coefficients of $\Delta_f^2 J_r(t)$, $\Delta_f\Sigma_f J_r(t)$,$\Delta_f\partial_{\Sigma_f}J_r$,$\Delta_f\partial_{\Delta_f}J_r$.

$$\Delta_f\partial_{\Delta_f}J_r\cdot\left(-\mathrm{i}\frac{d_1}{\dot{u}_1(t)}\right) = -\mathrm{i}\underbrace{[\frac{1}{2}\frac{d_1}{\dot{u}_1(t)}]}_{\Gamma(t)}(x-x')(\partial_x-\partial_{x'})J_r$$

$$\Delta_f\Sigma_f J_r\cdot\left(d_2 - \frac{d_1\dot{u}_2(t)}{\dot{u}_1(t)}\right) = \frac{1}{2}\underbrace{[d_2 - 2\Gamma(t)\dot{u}_2(t)]}_{\delta\Omega^2}(x^2-x'^2)J_r$$

$$\Delta_f\partial_{\Sigma_f}J_r\cdot\left(\frac{e_2+c_1}{\dot{u}_2(0)} + d_1\frac{2a_{21}(t)}{\dot{u}_1(t)\dot{u}_2(0)}\right) = 2\Gamma(t)\underbrace{\left(\frac{e_2+c_1}{2\Gamma(t)\dot{u}_2(0)} + \frac{2a_{21}(t)}{\dot{u}_2(0)}\right)}_{f(t)}(x-x')(\partial_x+\partial_{x'})J_r$$

$$\begin{aligned}
&\Delta_f^2 J_r\cdot\mathrm{i}\left(-(e_1+c_2) - (e_2+c_1)\frac{\dot{u}_2(t)}{\dot{u}_2(0)} - d_1\left(\frac{4a_{22}(t)}{\dot{u}_1(t)} + \frac{2a_{21}(t)}{\dot{u}_1(t)}\frac{\dot{u}_2(t)}{\dot{u}_2(0)}\right)\right)\\
&= -\mathrm{i}\left((e_1+c_2) + \frac{4a_{22}(t)d_1}{\dot{u}_1(t)} + \dot{u}_2(t)(\frac{e_2+c_1}{\dot{u}_2(0)} + d_1\frac{2a_{21}(t)}{\dot{u}_1(t)\dot{u}_2(0)})\right)(x-x')^2 J_r\\
&= -\mathrm{i}\Gamma(t)\underbrace{\left(\frac{e_1+c_2}{\Gamma(t)} + 8a_{22}(t) + 2\dot{u}_2(t)f(t)\right)}_{h(t)}(x-x')^2 J_r
\end{aligned}$$

1.3.8 主方程具体形式

By multiplying $\rho_r(x_i, x_i'; 0)$ with the reduced propagator J_r, we obtain the master equation in the coordinate representation.

$$\begin{aligned}
\mathrm{i}\partial_t\rho_r(x,x',t) &= -[\frac{1}{2M}(\partial_x^2-\partial_{x'}^2) + \frac{1}{2}M\Omega^2(x^2-x'^2)]\rho_r(x,x',t) + \frac{1}{2}\delta\Omega^2(x^2-x'^2)\rho_r(x,x',t)\\
&- \mathrm{i}\Gamma(t)(x-x')(\partial_x-\partial_{x'})\rho_r(x,x',t) + 2\Gamma(t)f(t)(x-x')(\partial_x+\partial_{x'})\rho_r(x,x',t) - \mathrm{i}\Gamma(t)h(t)(x-x')^2\rho_r(x,x',t)
\end{aligned} \tag{21}$$

In operator form:

$$\mathrm{i}\hbar\partial_t\hat{\rho} = [H_0+\delta H,\hat{\rho}] + \Gamma(t)[\hat{x},\{\hat{p},\hat{\rho}\}] - \mathrm{i}\Gamma(t)h(t)[\hat{x},[\hat{x},\hat{\rho}]] - \mathrm{i}2\Gamma(t)f(t)[\hat{x},[\hat{p},\hat{\rho}]] \tag{22}$$

We can see how obseverble moment evolution: $\frac{d}{dt}\langle A\rangle = \mathrm{Tr}(A\partial_t\rho)$, which lead to:

$$\frac{d}{dt}\langle p\rangle = -2\Gamma(t)\langle p\rangle - M\Omega_r^2\langle x\rangle$$

$$\frac{d}{dt}\langle p^2\rangle = 2\Gamma(t)h(t)\hbar + \cdots$$

$H_0+\delta H_0 = \frac{1}{2M}p^2 + \frac{1}{2}\Omega_r^2x^2$, $\frac{1}{\mathrm{i}\hbar}Tr(p,[H,\rho]) = -M\Omega_c^2\langle x\rangle$.

$$\frac{1}{\mathrm{i}\hbar}\mathrm{Tr}(A[x,\{p,\rho\}]) = \frac{1}{\mathrm{i}\hbar}\mathrm{Tr}([A,x]\cdot\{p,\rho\}),$$

$$\text{when } A=\hat{p},\quad \Gamma(t)\frac{1}{\mathrm{i}\hbar}\mathrm{Tr}(p[x,\{p,\rho\}]) = -2\langle p\rangle\Gamma(t)$$

So the second term in (22) is dissipation term. Which decrease the momentum of system.

$$\begin{aligned}
\frac{1}{\mathrm{i}\hbar}\operatorname{Tr}(A[x,[x,\rho]]) &= \frac{1}{\mathrm{i}\hbar}\operatorname{Tr}([A,x]\cdot[x,\rho]) \\
\text{when } A=\hat{p}^2,\quad \frac{1}{\mathrm{i}\hbar}\operatorname{Tr}(p^2[x,[x,\rho]]) &= \frac{1}{\mathrm{i}\hbar}\operatorname{Tr}([p^2,x][x,\rho]) \\
= \frac{1}{\mathrm{i}\hbar}\operatorname{Tr}(-2\mathrm{i}\hbar p[x,\rho]) &= \frac{1}{\mathrm{i}\hbar}\operatorname{Tr}(-2\mathrm{i}\hbar[p,x]\rho) = 2\mathrm{i}\hbar
\end{aligned}$$

这一项使得动量的二阶矩逐渐扩散。故为Diffuse term.

$$\begin{aligned}
\frac{1}{\mathrm{i}\hbar}\operatorname{Tr}(A[x,[p,\rho]]) &= \frac{1}{\mathrm{i}\hbar}\operatorname{Tr}([A,x]\cdot[p,\rho]) \\
\text{when } A=\hat{p}^2,\quad \frac{1}{\mathrm{i}\hbar}\operatorname{Tr}(p^2[x,[p,\rho]]) &= \frac{1}{\mathrm{i}\hbar}\operatorname{Tr}([p^2,x][p,\rho]) \\
= \frac{1}{\mathrm{i}\hbar}\operatorname{Tr}(-2\mathrm{i}\hbar p[p,\rho]) &= \frac{1}{\mathrm{i}\hbar}\operatorname{Tr}(-2\mathrm{i}\hbar[p,p]\rho) = 0
\end{aligned}$$

利用主方程计算纯度的变化$\frac{d}{dt}\langle\rho^2\rangle = 2\langle\rho\partial_t\rho\rangle$

$$\begin{aligned}
\frac{d}{dt}\langle\rho^2\rangle = 2\langle\rho\partial_t\rho\rangle &= 2\frac{1}{\mathrm{i}\hbar}\langle\rho([H,\hat{\rho}]+\Gamma(t)[\hat{x},\{\hat{p},\hat{\rho}\}]-\mathrm{i}\Gamma(t)h(t)[\hat{x},[\hat{x},\hat{\rho}]]-\mathrm{i}2\Gamma(t)f(t)[\hat{x},[\hat{p},\hat{\rho}]])\rangle \\
&= 2\Gamma(t)\operatorname{Tr}\rho^2 - 2\Gamma(t)h(t)\frac{1}{\hbar}\operatorname{Tr}([\rho,x]\cdot[x,\rho]) - \frac{4\Gamma(t)f(t)}{\hbar}\operatorname{Tr}(px\rho^2 - xp\rho^2 - 2\rho x\rho p).
\end{aligned}$$

1.3.9 Diffuse coefficient

Define:

$$D(t) = \Gamma(t)h(t) = \left((e_1+c_2) + \frac{4a_{22}(t)d_1}{\dot{u}_1(t)} - \dot{u}_2(t)(\frac{e_2+c_1}{\dot{u}_2(0)} + d_1\frac{2a_{21}(t)}{\dot{u}_1(t)\dot{u}_2(0)})\right) \tag{23}$$

$$e_1 = \int_0^t ds\nu(t-s)v_2(s), e_2 = \int ds\nu(t-s)v_1(s).$$

Extract the term

$$D_{main}(t) = \int_0^t ds\nu(t-s)\left(v_2(s) + \frac{\dot{u}_2(t)}{\dot{u}_2(0)}v_1(s)\right).$$

2 Influence functionals in scalar field models

The main difference is that the interaction is no longer linear, but instead takes the quartic form ϕ^4. This leads to several types of terms, such as ϕ_S^4, $\phi_S^3\phi_F$, $\phi_S^2\phi_F^2$, $\phi_S\phi_F^3$, and ϕ_F^4. We can analyze these contributions term by term.

2.1 $\phi_S^3\phi_F$ term

Among them, the term $\phi_S^3\phi_F$ is the most similar to the oscillator case we previously discussed. If we simplify the slow sector to a single mode, we denote it by

$$x = \sqrt{\frac{\hbar}{2m\omega_S}}(a+a^\dagger).$$

We then consider its interaction with the fast modes

$$Q = \sum_{i,\omega_i>b\Lambda}\sqrt{\frac{\hbar}{2m\omega_i}}(b_i+b_i^\dagger),$$

through a coupling of the form $\lambda x^3 Q$. When we consider this kind of interaction, I want to see the difference in master equation between the linear case.

The correlation function of Q is same with (6).

The influence functional is almost same with (7).

$$\delta A[x,x'] = \mathrm{i}\int_0^t ds_1 \int_0^{s_1} ds_2 \Delta_3(s_1)\nu(s_1-s_2)\Delta_3(s_2) - 2\int_0^t ds_1 \int_0^{s_1} ds_2 \Delta_3(s_1)\eta(s_1-s_2)\Sigma_3(s_2)$$
$$\text{where} \quad \Delta_3(s) = x^3(s) - x'^3(s), \Sigma_3(s) = \frac{x^3(s)+x'^3(s)}{2}. \tag{24}$$

The calculation is still same in reduced propagator (12), except change the coordinate Σ, Δ to Σ_3, Δ_3.

$$J_r(t+\Delta t; x_f, x'_f) = N(t)\int_0^\infty dx_m \int_0^\infty dx'_m \exp\left\{[\mathrm{i}\frac{M}{2\Delta t}((x_f-x_m)^2-(x'_f-x'_m)^2)]\right\}\cdot\left[1-\mathrm{i}\Delta t V(x_f)-\mathrm{i}\Delta t V(x'_f)\right]$$
$$\cdot\int_i^m \mathcal{D}\bar{x}\mathcal{D}\bar{x}'\exp\left\{[\mathrm{i}\left(S[\bar{x}]-S[\bar{x}']-2\int_0^t ds_1\int_0^{s_1} ds_2\bar{\Delta}_3(s_1)\eta(s_1-s_2)\bar{\Sigma}_3(s_2)+\mathrm{i}\int_0^t ds_1\int_0^{s_1} ds_2\bar{\Delta}_3(s_1)\nu(s_1-s_2)\bar{\Delta}_3(s_2)\right)]\right\}$$
$$\cdot\exp\left\{\mathrm{i}\Delta t\left[-2\int_0^t ds\bar{\Sigma}_3(s)J_{\Sigma_3}(s)+\mathrm{i}\int_0^t ds\bar{\Delta}_3(s)J_{\Delta_3}(s)\right]\right\}$$
$$\text{where} \quad J_{\Sigma_3}(s) = \frac{1}{\Delta t}\int_t^{t+\Delta t} ds_1\tilde{\Delta}_3(s_1)\eta(s_1-s), J_{\Delta_3}(s) = \frac{1}{\Delta t}\int_t^{t+\Delta t} ds_1\tilde{\Delta}_3(s_1)\nu(s_1-s). \tag{25}$$

Approximate with first-order accuracy in Δt, $J_{\Sigma_3}(s) \approx \Delta_{3,f}\eta(t-s), J_{\Delta_3} \approx \Delta_{3,f}\nu(t-s)$.

Although Δ_3 and Σ_3 appear in δA, the main part $S[x]-S[x']$ is still expressed in terms of Δ and Σ.

2.1.1 鞍点

In (12), we can express the Δ_3/Σ_3 terms to some functional of Δ and Σ. When evaluated the saddle equation of Δ_{cl}/Σ_{cl}, we use follow formulas.

$$\text{Assume } I(\Delta,\Sigma) = -2\int_0^t ds_1\int_0^{s_1} ds_2 f(\Delta(s_1),\Sigma(s_1))\eta(s_1-s_2)g(\Delta(s_2),\Sigma(s_2)),$$
$$\frac{\delta I}{\delta\Delta(s)} = -2\left[(\partial_\Delta f)(s)\int_0^s ds'\,\eta(s-s')g(s') + (\partial_\Delta g)(s)\int_s^t ds' f(s')\eta(s'-s)\right], \tag{26}$$
$$\frac{\delta I}{\delta\Sigma(s)} = -2\left[(\partial_\Sigma f)(s)\int_0^s ds'\,\eta(s-s')g(s') + (\partial_\Sigma g)(s)\int_s^t ds' f(s')\eta(s'-s)\right].$$

$$\Delta_3 = x^3 - x'^3 = (\Sigma+\frac{\Delta}{2})^3 - (\Sigma-\frac{\Delta}{2})^3 = 3\Sigma^2\Delta + \frac{\Delta^3}{4},$$
$$\Sigma_3 = \frac{1}{2}((\Sigma+\frac{\Delta}{2})^3 + (\Sigma-\frac{\Delta}{2})^3) = \Sigma^3 + \frac{3}{4}\Sigma\Delta^2;$$
$$\partial_\Delta\Delta_3 = 3\Sigma^2 + \frac{3}{4}\Delta^2, \partial_\Sigma\Delta_3 = 6\Sigma\Delta; \tag{27}$$
$$\partial_\Delta\Sigma_3 = \frac{3}{2}\Sigma\Delta, \partial_\Sigma\Sigma_3 = 3\Sigma^2 + \frac{3}{4}\Delta^2.$$

The saddle equation change to:

$$\ddot{\Sigma}_{\rm cl}(s)+\Omega^2\Sigma_{\rm cl}(s)+2\left[(\partial_\Delta\Delta_{3,cl})(s)\int_0^s ds'\,\eta(s-s')\Sigma_{3,cl}(s')+(\partial_\Delta\Sigma_{3,cl})(s)\int_s^t ds'\Delta_{3,cl}(s')\eta(s'-s)\right]=0, \tag{28}$$

$$\ddot{\Delta}_{\rm cl}(s)+\Omega^2\Delta_{\rm cl}(s)+2\ \left[(\partial_\Sigma\Delta_{3,cl})(s)\int_0^s ds'\,\eta(s-s')\Sigma_{3,cl}(s')+(\partial_\Sigma\Sigma_{3,cl})(s)\int_s^t ds'\Delta_{3,cl}(s')\Delta_{3,cl}(s'-s)\right]=0, \tag{29}$$

If we expand coordinates around saddle points $\Sigma=\Sigma_{cl}+\chi_1,\Delta=\Delta_{cl}+\chi_2$, the collective coordinates have expression :$\Delta_3=\Delta_{3,cl}+\frac{\partial\Delta_3}{\partial\Sigma}|_{cl}\chi_1+\frac{\partial\Delta_3}{\partial\Delta}|_{cl}\chi_2+\cdots$; $\Sigma_3=\Sigma_{3,cl}+\frac{\partial\Sigma_3}{\partial\Sigma}|_{cl}\chi_1+\frac{\partial\Sigma_3}{\partial\Delta}|_{cl}\chi_2+\cdots$. Similar to (15) the reduced propagator become:

$$\begin{aligned}A_0[\chi_1,\chi_2]=A_0[\Delta_{cl},\Sigma_{cl}]-2\int_0^t ds_1\int_0^{s_1}ds_2(\Delta_{3,cl}+\partial_\Sigma\Delta_3\chi_1+\partial_\Delta\Delta_3\chi_2)_{s_1}\eta(s_1-s_2)(\Sigma_{3,cl}+\partial_\Sigma\Sigma_3\chi_1+\partial_\Delta\Sigma_3\chi_2)_{s_2}\\ +{\rm i}\int_0^t ds_1\int_0^{s_1}ds_2(\Delta_{3,cl}+\partial_\Sigma\Delta_3\chi_1+\partial_\Delta\Delta_3\chi_2)_{s_1}\nu(s_1-s_2)(\Delta_{3,cl}+\partial_\Sigma\Delta_3\chi_1+\partial_\Delta\Delta_3\chi_2)_{s_2}\end{aligned}$$

$$\delta A[\chi_1,\chi_2]=\delta A[\Delta_{cl},\Sigma_{cl}]+\Delta t\left\{-2\int_0^t ds(\partial_\Sigma\Sigma_3(s)\chi_1(s)+\partial_\Delta\Sigma_3(s)\chi_2(s))J_{\Sigma_3}+{\rm i}\int_0^t ds(\partial_\Sigma\Delta_3(s)\chi_1(s)+\partial_\Delta\Delta_3(s)\chi_2(s))J_{\Delta_3}\right\} \tag{30}$$

$$J_r[m;i;J_{\Sigma,\Delta}]=\exp\left\{{\rm i}(A_0[\Sigma_{cl},\Delta_{cl}]+\delta A[\Sigma_{cl},\Delta_{cl}])\right\}\underbrace{\int_0^0\mathcal{D}\chi_1\int_0^0\mathcal{D}\chi_2{\rm quadratic\ form\ of}\chi}_{J_r[0,0,t|0,0;{\rm source}]} \tag{31}$$

$$J_r[0,0,t|0,0;{\rm source}]=\int_0^0\mathcal{D}\chi_1\int_0^0\mathcal{D}\chi_2\exp\left\{{\rm i}\left[\int_0^t ds_1\int_0^t ds_2\frac{1}{2}\chi_i(s_1)\mathcal{O}_{ij}(s_1,s_2)\chi_i(s_2)+\int_0^t ds\chi_iB_i\right]\right\}$$

$$\begin{aligned}O_{11}(s_1,s_2)&=2i\,\nu(s_1,s_2)\partial_\Sigma\Delta_3(s_1)\partial_\Sigma\Delta_3(s_2)-4\eta(s_1,s_2)\partial_\Sigma\Delta_3(s_1)\partial_\Sigma\Sigma_3(s_1)\theta(s_1-s_2),\\ O_{22}(s_1,s_2)&=2i\,\nu(s_1,s_2)\partial_\Delta\Delta_3(s_1)\partial_\Delta\Delta_3(s_2)-4\eta(s_1,s_2)\partial_\Delta\Delta_3(s_1)\partial_\Delta\Sigma_3(s_1)\theta(s_1-s_2),\end{aligned} \tag{32}$$

$$O_{12}(s_1,s_2)=\left[\frac{d^2}{ds_1^2}+\Omega^2\right]\delta(s_1-s_2)+2\theta(s_2-s_1)\,\eta(s_2-s_1)\partial_\Delta\Delta_3(s_1)\partial_\Sigma\Sigma_3(s_2)+2\theta(s_1-s_2)\,\eta(s_1-s_2)\partial_\Sigma\Delta_3(s_1)\partial_\Delta\Sigma_3(s_2),$$

$$O_{21}(s_1,s_2)=\left[\frac{d^2}{ds_1^2}+\Omega^2\right]\delta(s_1-s_2)+2\theta(s_1-s_2)\,\eta(s_1-s_2)\partial_\Delta\Delta_3(s_1)\partial_\Sigma\Sigma_3(s_2)+2\theta(s_2-s_1)\,\eta(s_2-s_1)\partial_\Sigma\Delta_3(s_1)\partial_\Delta\Sigma_3(s_2).$$

$$\begin{aligned}B_1(s)&=-2\Delta tJ_\Sigma(s)\partial_\Sigma\Sigma_3+{\rm i}\Delta tJ_{\Delta_3}\partial_\Sigma\Delta_3+{\rm i}\tilde{J}_{\Delta_3}\partial_\Sigma\Delta_3,\\ B_2(s)&=-2\Delta tJ_\Sigma(s)\partial_\Delta\Sigma_3+{\rm i}\Delta tJ_{\Delta_3}(s)\partial_\Delta\Delta_3+{\rm i}\tilde{J}_{\Delta_3}\partial_\Delta\Delta_3.\end{aligned}$$

$$J_r[0,0,t|0,0;\boldsymbol{B}]=\mathcal{N}\exp\left\{-{\rm i}\frac{1}{2}\int_0^t ds\int_0^t ds'B_i(s)G_{ij}(s,s')B_j(s')\right\} \tag{33}$$

$$J_r[m;i;J_{\Sigma,\Delta}] = \underbrace{\mathcal{N}\exp\{\mathrm{i}(A_0[\Sigma_{cl},\Delta_{cl}])\}}_{\approx J_r[x_m,x'_m,t;x_i,x'_i]}$$

$$\cdot\exp\left\{\mathrm{i}\Delta t\left[-2\int_0^t ds\Sigma_{3cl}(s)J_{\Sigma_3}(s)+\mathrm{i}\int_0^t ds\Delta_{3cl}(s)J_{\Delta_3}(s)\right]-\mathrm{i}\frac{1}{2}\int_0^t ds\int_0^t ds'\underbrace{B_i(s)G_{ij}(s,s')B_j(s')}_{\text{just keep in }\Delta t\text{ order.}}\right\}$$

Compare to (15), this case is more complicate because of the mix due to the dependence of Σ/Δ in Σ_3 and Δ_3. The procedure used to derive the correct master equation in the linear case is much harder to implement in the case of cubic interactions.

$$J_r(t+\Delta t;x,x') = J_r(t;x,x')+\frac{\mathrm{i}}{2}(\partial_x^2 J_r)\frac{\Delta t}{M}-\frac{i}{2}(\partial_{x'}^2 J_r)\frac{\Delta t}{M}+J_r(t,;x,x')\Delta t\cdot$$
$$\left(-\mathrm{i}V(x)+\mathrm{i}V(x')+\mathrm{i}\left[-2\int_0^t ds\Sigma_{3cl}(s)J_{\Sigma_3}(s)+\mathrm{i}\int_0^t ds\Delta_{3cl}(s)J_{\Delta_3}(s)-\frac{1}{2}\int_0^t ds\int_0^t ds'\underbrace{B_i(s)G_{ij}(s,s')B_j(s')}_{\text{just keep in }\Delta t\text{ order.}}\right]\right) \tag{34}$$

2.1.2 处理初始条件相关变量Σ_i,Δ_i

This step need to evaluate $J_r(t,f;i,0)$. In our saddle point choice, the expression is:

$$J_r(t,f;i,0) = \mathcal{N}\exp\left\{\mathrm{i}M(\Delta_f\dot{\Sigma}_f-\Delta_i\dot{\Sigma}_i)-\frac{1}{2}\int \mathrm{d}s_1\int \mathrm{d}s_2\Delta_{3,cl}(s_1)\nu(s_1-s_2)\Delta_{3,cl}(s_2)\right\} \tag{35}$$

$\Sigma_{cl}=\Sigma_i u_1(s)+\Sigma_f u_2(s);\Delta_{cl}=\Delta_i v_1(s)+\Delta_f v_2(s)$.

$$\begin{aligned}\Delta_{3,cl}(s_1) &= 3\Sigma_i^2\Delta_i u_1^2 v_1+3\Sigma_i^2\Delta_f u_1^2 v_2+3\Sigma_f^2\Delta_i u_2^2 v_1\\ &+3\Sigma_f^2\Delta_f u_2^2 v_2+6\Sigma_i\Sigma_f\Delta_i u_1 u_2 v_1+6\Sigma_i\Sigma_f\Delta_f u_1 u_2 v_2\\ &+\frac{\Delta_i^3}{4}v_1^3+\frac{3}{4}\Delta_i^2\Delta_f v_1^2 v_2+\frac{3}{4}\Delta_i\Delta_f^2 v_1 v_2^2+\frac{\Delta_f^3}{4}v_2^3\end{aligned} \tag{36}$$

$\Delta_{3,cl}(t)=3\Sigma_f^2\Delta_f+\frac{\Delta_f^3}{4}\equiv\Delta_{3,f}$; and similar formula would apply to $\Delta_{3,cl}(s)$. In the exponential, there are total $(4\times4\times4)^2$ term related to $\Sigma_i,\Sigma_f,\Delta_i,\Delta_f$. I can just write the term related to the main part.

$$\begin{cases}\Delta_i J_r = \left[\frac{\dot{u}_2(t)}{\dot{u}_2(0)}\Delta_f+\mathrm{i}\frac{1}{\dot{u}_2(0)}\partial_{\Sigma_f}+(\text{tedious terms with integral of kernel }\nu)\right]J_r\\ \Sigma_i J_r(t,0) = \left[-\frac{\dot{u}_2(t)}{\dot{u}_1(t)}\Sigma_f-\mathrm{i}\frac{1}{\dot{u}_1(t)}\frac{\partial}{\partial\Delta_f}+(\text{tedious terms with integral of kernel }\nu)\right]J_r(t,0)\end{cases} \tag{37}$$

In the following calculation, we treat Δ_i as $\frac{\dot{u}_2(t)}{\dot{u}_2(0)}\Delta_f$, and ignore the corresponding contribution due to its complexity [1].

2.1.3 term related to diffuse

I just pick up term $\Delta_{3,f}^2 J_r\sim(x^3-x'^3)^2 J_r\sim[\hat{x}^3,[\hat{x}^3,\hat{\rho}]]$, because I know this term would produce diffuse and decoherence.

The term $-[\int_0^t \mathrm{d}s\Delta_{3cl}(s)J_{\Delta_3}(s)]J_r$ in (34) would give a contribution, which similar to the e_i term in (22). [2]

$$(-)[\int_0^t \mathrm{d}s\Delta_{3,cl}(s)\nu(t-s)]\Delta_{3,f}J_r$$

[1] These term are order λ^2

[2] Because I didn't treat the term related to G_{ij}, so there is no term related to c_i occur in master equation, which actually did.

$$\Delta_{3cl}(s) = 3\Sigma_f^2\Delta_f\left(u_2(s) - \frac{\dot{u}_2(t)}{\dot{u}_1(t)}u_1(s)\right)^2\left(\frac{\dot{u}_2(t)}{\dot{u}_2(0)}v_1(s) + v_2(s)\right) + \frac{\Delta_f^3}{4}\left(\frac{\dot{u}_2(t)}{\dot{u}_2(0)}v_1(s) + v_2(s)\right)^3 \tag{38}$$

$$\begin{aligned}\Delta_{3cl}(s) =& \Delta_{3,f}\left(u_2(s) - \frac{\dot{u}_2(t)}{\dot{u}_1(t)}u_1(s)\right)^2\left(\frac{\dot{u}_2(t)}{\dot{u}_2(0)}v_1(s) + v_2(s)\right)\\ &+ \frac{\Delta_f^3}{4}\left(\frac{\dot{u}_2(t)}{\dot{u}_2(0)}v_1(s) + v_2(s)\right)\left[\left(\frac{\dot{u}_2(t)}{\dot{u}_2(0)}v_1(s) + v_2(s)\right)^2 - \left(u_2(s) - \frac{\dot{u}_2(t)}{\dot{u}_1(t)}u_1(s)\right)^2\right]\end{aligned} \tag{39}$$

Define some coefficients:

$$\begin{aligned}D_1(t) &= \int_0^t\left(u_2(s) - \frac{\dot{u}_2(t)}{\dot{u}_1(t)}u_1(s)\right)^2\left(\frac{\dot{u}_2(t)}{\dot{u}_2(0)}v_1(s) + v_2(s)\right)\nu(t-s)\mathrm{d}s\\ D_2(t) &= \frac{1}{4}\int_0^t\left(\frac{\dot{u}_2(t)}{\dot{u}_2(0)}v_1(s) + v_2(s)\right)\left[\left(\frac{\dot{u}_2(t)}{\dot{u}_2(0)}v_1(s) + v_2(s)\right)^2 - \left(u_2(s) - \frac{\dot{u}_2(t)}{\dot{u}_1(t)}u_1(s)\right)^2\right]\nu(t-s)\mathrm{d}s\end{aligned} \tag{40}$$

D_1 correspond to term $\Delta_{3,f}^2 J_r$,and D_2 correspond to term $\Delta_f^3\Delta_{3,f}J_r$. Multiply with $\rho_r(x_i, x_i'; 0)$, we could get part of master equation:

$$\partial_r\rho_r(x,x';t) = \frac{\mathrm{i}}{2M}(\partial_x^2 - \partial_{x'}^2)\rho_r(t) - \mathrm{i}(V(x) - V(x'))\rho_r(t) - D_1(t)(x^3 - x'^3)^2\rho_r(x,x';t) - D_2(t)(x-x')^3(x^3 - x'^3)\rho_r(x,x';t) \tag{41}$$

In operator form:

$$\partial_t\hat{\rho}(t) = -\mathrm{i}[H,\hat{\rho}] - D_1(t)[\hat{x}^3,[\hat{x}^3,\hat{\rho}]] - D_2(t)[\hat{x},[\hat{x},[\hat{x}^3,\hat{\rho}]]] + \cdots \tag{42}$$

2.2 $\phi_S\phi_F^3$ term

Assume the interaction is $H_I = \sum_n \lambda x q_n^3$. The main difference in this case is the correlation function(spectrum density) of bath degree of freedom.

Define $Q_3 = \lambda\sum_n q_n^3$, the influence functional can be calculate as follow:

$$\begin{aligned}\delta A[x,x'] = \int_0^t ds[x(s)\langle Q_3(s)\rangle - x'(s)\langle Q_3'(s)\rangle] + \frac{\mathrm{i}}{2}\int ds\int ds' x(s)\langle\mathcal{T}Q_3(s)Q_3(s')\rangle x(s') + x'(s)\langle\bar{\mathcal{T}}Q_3(s)Q_3(s')\rangle x'(s')\\ -x(s)\langle Q_3(s')Q_3(s)\rangle x'(s') - x(s')\langle Q_3(s)Q_3(s')\rangle x'(s) + \cdots\end{aligned} \tag{43}$$

$$\begin{aligned}\langle Q_3(s)Q_3(s')\rangle &= \lambda^2\sum_n\left(6\langle q_n(s)q_n(s')\rangle^3 + 9\langle q_n(s)q_n(s')\rangle\langle q_n^2\rangle^2\right)\\ &= \lambda^2\sum_n\left(6\nu_n^3(s-s') - 6\nu_n(s-s')\eta^2(s-s') + 9\langle q_n^2\rangle^2\nu_n(s-s')\right)\\ &+\mathrm{i}\lambda^2\sum_n\left(6\nu_n^2(s-s')\eta_n(s-s') - 9\eta_n^3(s-s') + 9\langle q_n^2\rangle^2\eta_n(s-s')\right)\\ &= \nu_3(s-s') + \mathrm{i}\eta_3(s-s')\end{aligned} \tag{44}$$

The influence functional is almost same with (7).

$$\delta A[x,x'] = \mathrm{i}\int_0^t ds_1\int_0^{s_1} ds_2\Delta(s_1)\nu_3(s_1-s_2)\Delta(s_2) - 2\int_0^t ds_1\int_0^{s_1} ds_2\Delta(s_1)\eta_3(s_1-s_2)\Sigma(s_2) \tag{45}$$

The noise kernel ν_3 is even and the dissipation kernel η_3 is odd, just as in the case of ν and η. Therefore, the calculation proceeds in the same way as in the linear interaction case, except that ν and η are replaced by ν_3 and η_3.

2.3 推广到场

To begin with, the action of scalar fields we study is below[3] [4]:

$$
\begin{aligned}
S[\phi_{S/F}] &= \int \mathrm{d}^4x \left[\frac{1}{2}\partial_\mu\phi_{S/F}(x)\partial^\mu\phi_{S/F}(x) - \frac{1}{2}m^2\phi_{S/F}^2(x)\right] \\
\phi_S(x) = \int_{|\vec{k}|<b\Lambda} \frac{\mathrm{d}^3\boldsymbol{k}}{(2\pi)^3}\phi(\vec{k},t)e^{\mathrm{i}\boldsymbol{x}\cdot\boldsymbol{k}}, \phi_F(x) &= \int_{b\Lambda<|\vec{k}|<\Lambda} \frac{\mathrm{d}^3\boldsymbol{k}}{(2\pi)^3}\phi(\vec{k},t)e^{\mathrm{i}\boldsymbol{x}\cdot\boldsymbol{k}}; \\
S_{int} &= -\int \mathrm{d}^4x \left\{\frac{\lambda}{4!}(\phi_S^4+\phi_F^4) + \frac{\lambda}{4}\phi_S^2\phi_F^2 + \frac{\lambda}{6}\phi_S^3\phi_F + \frac{\lambda}{6}\phi_S\phi_F^3\right\}.
\end{aligned}
\tag{46}
$$

First we have figured out the infulence fuctional, and I use ϕ to express ϕ_S. Use $J(\phi)O(\phi_F)$ to describe the interaction action. There exist 3 kinds of interaction $J_1(\phi)O_3(\phi_F) = \frac{1}{6}\lambda\phi\phi_F^3$ and so on.

$$
\begin{aligned}
\rho_r(\phi_f(\boldsymbol{x}),\phi_f'(\boldsymbol{x}),t) &= \int \mathcal{D}\phi_i()\mathcal{D}\phi_i'()J_r[\phi_f(\boldsymbol{x}),\phi_f'(\boldsymbol{x}),t;\phi_i,\phi_i',0]\rho_r(\phi_i(\boldsymbol{x}),\phi_i'(\boldsymbol{x}),0), \\
J_r[\phi_f(\boldsymbol{x}),\phi_f'(\boldsymbol{x}),t;\phi_i,\phi_i',0] &= \int_{0,\phi_i(\boldsymbol{x})}^{t,\phi_f(\boldsymbol{x})} \mathcal{D}\phi(x)\mathcal{D}\phi'(x)e^{\mathrm{i}S[\phi]-\mathrm{i}S[\phi']}e^{\mathrm{i}\delta A[\phi(t),\phi'(t)]}
\end{aligned}
\tag{47}
$$

$e^{\mathrm{i}\delta A} = \langle e^{\mathrm{i}(S_{int}(\phi;\phi_F)-S_{int}(\phi',\phi_F'))}\rangle_{\phi_F,\phi_F'}$,

$$
\begin{aligned}
\delta A[\phi(x),\phi(s')] = &-\int \mathrm{d}^4x(J[\phi(x)]-J[\phi'(x)])\langle O(x)\rangle \\
&+\frac{\mathrm{i}}{2}\int \mathrm{d}^4x_1\int \mathrm{d}^4x_2 J(x_1)\langle\mathcal{T}O(x_1)O(x_2)\rangle J(x_2) + J'(x_1)\langle\bar{\mathcal{T}}O(x_1)O(x_2)\rangle)J'(x_2) - J(x_1)\langle O(x_1)O(x_2)\rangle J'(x_2) \\
&-J'(x_1)\langle O(x_1)O(x_2)\rangle J(x_2) - J(x_1)\langle O(x_1)\rangle\langle O(x_2)\rangle J(x_2) - J'(x_1)\langle O(x_1)\rangle\langle O(x_2)\rangle J'(x_2) \\
&+J(x_1)\langle O(x_1)\rangle\langle O(x_2)\rangle J'(x_2) + J'(x_1)\langle O(x_1)\rangle\langle O(x_2)\rangle J(x_2).
\end{aligned}
\tag{48}
$$

$$
\begin{aligned}
\delta A[\phi(x),\phi(s')] = &-\int \mathrm{d}^4x(J[\phi(x)]-J[\phi'(x)])\langle O(x)\rangle \\
&+\frac{\mathrm{i}}{2}\int \mathrm{d}^4x_1\int \mathrm{d}^4x_2 J(x_1)\langle\mathcal{T}O(x_1)O(x_2)\rangle_c J(x_2) + J'(x_1)\langle\bar{\mathcal{T}}O(x_1)O(x_2)\rangle_c J'(x_2) - J(x_1)\langle O(x_1)O(x_2)\rangle_c J'(x_2) \\
&-J'(x_1)\langle O(x_1)O(x_2)\rangle_c J(x_2)
\end{aligned}
$$

Define Wightman function: $W_+(x_1-x_2) = \langle O(x_1)O(x_2)\rangle - \langle O(x_1)\rangle\langle O(x_2)\rangle$, and $W_-(x_1-x_2) = \langle O(x_2)O(x_1)\rangle - \langle O(x_1)\rangle\langle O(x_2)\rangle$. They are conjugate to each other $W_+^*(x_1-x_2) = W_-(x_1-x_2)$. Define:

$$
\nu(x_1-x_2) = \mathrm{Re}\,W_+(x_1-x_2); \eta(x_1-x_2) = \mathrm{Im}\,W_+(x_1-x_2). \tag{49}
$$

The explicitly form of Wightman function is follow:

$$
\begin{aligned}
\delta A[\phi,\phi'] &= \mathrm{i}\int_0^t \mathrm{d}^4x_1\int_0^{s_1} \mathrm{d}^4x_2(J(x_1)-J'(x_1))\nu(x_1-x_2)(J(x_2)-J'(x_2)) + \mathrm{i}(J(x_1)-J'(x_2))\eta(x_1-x_2)(J(x_2)+J'(x_2)) \\
&= \mathrm{i}\int_0^t \mathrm{d}^4x_1\int_0^{s_1} \mathrm{d}^4x_2(\Delta_J(x_1)\nu(x_1-x_2)\Delta_J(x_2) - 2\int_0^t \mathrm{d}^4x_1\int_0^{s_1} \mathrm{d}^4x_2(\Delta_J(x_1)\eta(x_1-x_2)\Sigma_J(x_2) \\
&\text{where} \quad \Delta_J(x) = J[\phi(x)]-J[\phi'(x)], \Sigma_J(x) = \frac{J[\phi(x)]+J[\phi'(x)]}{2}.
\end{aligned}
\tag{50}
$$

[3] the time integral range is 0 to t, because we consider time evolution

[4] Represent three-dimensional vectors using bold letters

Then, we evaluate reduced propagator at time $t+\Delta t$, $J_r[\phi_f,\phi'_f;t+\Delta t]$

插入中间场，$\phi_m,\phi'_m.\tilde{\phi}(\boldsymbol{x},s)=\frac{\phi_f(\boldsymbol{x})-\phi_m(\boldsymbol{x})}{\Delta t}\cdot(s-t)+\phi_m(\boldsymbol{x})$.

$$
\begin{aligned}
J_r[\phi_f,\phi'_f,t+\Delta t]=\int\mathcal{D}\phi_m(\boldsymbol{x})\int\mathcal{D}\phi'_m(\boldsymbol{x})\exp\left\{\mathrm{i}\int\mathrm{d}^3\boldsymbol{x}\frac{1}{2\Delta t}\left[(\phi_f(\boldsymbol{x})-\phi_m(\boldsymbol{x}))^2-(\phi'_f(\boldsymbol{x})-\phi'_m(\boldsymbol{x}))^2\right]\right\}\\
\cdot\left(1-\mathrm{i}\int\mathrm{d}\boldsymbol{x}\frac{\Delta t}{2}[(\nabla\phi_m(\boldsymbol{x}))^2+m^2\phi_m^2-(\nabla\phi'_m)^2-m^2\phi_m'^2]\right)\\
\cdot\int\mathcal{D}\bar{\phi}(x)\mathcal{D}\bar{\phi}'(x)\exp\left\{\mathrm{i}(S[\bar{\phi}]-S[\bar{\phi}'])\right\}\exp\left\{\mathrm{i}\delta A[\bar{\phi},\bar{\phi}';\tilde{\phi},\tilde{\phi}']\right\}
\end{aligned}
\tag{51}
$$

For convenience, I call $t_{x_1}=s_1$ and $t_{x_2}=s_2$,$t_x=s$.

$$
\begin{aligned}
\delta A[\bar{\phi},\bar{\phi}';\tilde{\phi},\tilde{\phi}']=-2\int_{\boldsymbol{x}_1}\int_{\boldsymbol{x}_2}\int_0^t\mathrm{d}s_1\int_0^{s_1}\mathrm{d}s_2\bar{\Delta}_J(\boldsymbol{x}_1,s_1)\eta(x_1-x_2)\bar{\Sigma}_J(\boldsymbol{x}_2,s_2)\\
+\mathrm{i}\int_0^t\mathrm{d}^4x_1\int_0^{s_1}\mathrm{d}^4x_2\bar{\Delta}_J(x_1)\nu(x_1-x_2)\bar{\Delta}_J(x_2)\\
+\Delta t\left[-2\int_0^t\mathrm{d}^4x\bar{\Sigma}_J(x)J_\Sigma(x)+\mathrm{i}\int_0^t\mathrm{d}^4x\bar{\Delta}_J(x)J_\Delta(x)\right]
\end{aligned}
\tag{52}
$$

The define of the currents are $J_\Sigma(x)=\frac{1}{\Delta t}\int_t^{t+\Delta t}\mathrm{d}x_1\tilde{\Delta}_J(x_1)\eta(x_1-x)$, and $J_\Delta(x)=\frac{1}{\Delta t}\int_t^{t+\Delta t}\mathrm{d}x_1\tilde{\Delta}_J(x_1)\nu(x_1-x)$. Approximate with first-order accuracy in Δt, $J_\Sigma(\boldsymbol{x},s)\approx\int\mathrm{d}\boldsymbol{x}_1\Delta_{J,f}(\boldsymbol{x}_1)\eta(\boldsymbol{x}_1-\boldsymbol{x},t-s)$, $J_\Delta(\boldsymbol{x},s)\approx\int\mathrm{d}\boldsymbol{x}_1\Delta_{J,f}(\boldsymbol{x}_1)\nu(\boldsymbol{x}_1-\boldsymbol{x},t-s)$.

$$
\begin{aligned}
A_0[\Sigma,\Delta]=\int\mathrm{d}^4x(\partial_\mu\Delta(x)\partial^\mu\Sigma(x)-m^2\Delta(x)\Sigma(x))\\
-2\int_0^t\mathrm{d}^4x_1\int_0^{s_1}\mathrm{d}^4x_2\,\Delta_J(x_1)\eta(x_1-x_2)\Sigma_J(x_2)\\
+\mathrm{i}\int_0^t\mathrm{d}^4x_1\int_0^{s_1}\mathrm{d}^4x_2\,\Delta_J(x_1)\nu(x_1-x_2)\Delta_J(x_2)
\end{aligned}
\tag{53}
$$

where the $\Delta(x)=\phi(x)-\phi'(x)$,and $\Sigma(x)=\frac{\phi(x)+\phi'(x)}{2}$, which is defferent with $\Delta_J(x)$.

We can express the Δ_J/Σ_J in terms of the original fields $\Delta/\Sigma(x)$ like the case of cubic interaction $\phi_S^3\phi_F$.

For the four-dimensional generalization of (26), let

$$
I[\Delta,\Sigma]=-2\int_0^t\mathrm{d}^4x_1\int_0^{s_1}\mathrm{d}^4x_2\,f(\Delta(x_1),\Sigma(x_1))\eta(x_1-x_2)g(\Delta(x_2),\Sigma(x_2)),
$$

where $x_i=(t_i,\boldsymbol{x}_i)$ and $\mathrm{d}^4x_i=\mathrm{d}t_i\,\mathrm{d}^3\boldsymbol{x}_i$. Then

$$
\begin{aligned}
\frac{\delta I}{\delta\Delta(x)}&=-2\left[(\partial_\Delta f)(x)\int_0^s\mathrm{d}^4x'\,\eta(x-x')g(x')+(\partial_\Delta g)(x)\int_s^t\mathrm{d}^4x'\,f(x')\eta(x'-x)\right],\\
\frac{\delta I}{\delta\Sigma(x)}&=-2\left[(\partial_\Sigma f)(x)\int_0^s\mathrm{d}^4x'\,\eta(x-x')g(x')+(\partial_\Sigma g)(x)\int_s^t\mathrm{d}^4x'\,f(x')\eta(x'-x)\right].
\end{aligned}
$$

The saddle equations are then generalized to the field case as

$$\left(\partial_\mu\partial^\mu + m^2\right)\Sigma_{\rm cl}(x) + 2\left[(\partial_\Delta\Delta_{J,\rm cl})(x)\int_0^s \mathrm{d}^4x_2\,\eta(x-x_2)\Sigma_{J,\rm cl}(x_2) + (\partial_\Delta\Sigma_{J,\rm cl})(x)\int_s^t \mathrm{d}^4x_2\,\Delta_{J,\rm cl}(x_2)\eta(x_2-x)\right] = 0, \tag{54}$$

$$\left(\partial_\mu\partial^\mu + m^2\right)\Delta_{\rm cl}(x) + 2\left[(\partial_\Sigma\Delta_{J,\rm cl})(x)\int_0^s \mathrm{d}^4x_2\,\eta(x-x_2)\Sigma_{J,\rm cl}(x_2) + (\partial_\Sigma\Sigma_{J,\rm cl})(x)\int_s^t \mathrm{d}^4x_2\,\Delta_{J,\rm cl}(x_2)\eta(x_2-x)\right] = 0, \tag{55}$$

where $\Delta_{J,\rm cl} = J[\phi_{\rm cl}] - J[\phi'_{\rm cl}]$.

A Thermal average

$$q_n(t) = \sqrt{\frac{\hbar}{2m\omega_n}}(b_n e^{-\mathrm{i}\omega_n t} + b_n^\dagger e^{\mathrm{i}\omega_n t}),$$

$$\langle b\rangle_\beta = 0 \quad , \langle b^\dagger\rangle_\beta = 0.$$

$$\langle b_n^\dagger b_n\rangle_\beta = \frac{1}{e^{\hbar\beta\omega_n}-1}, \langle b_n b_n^\dagger\rangle_\beta = \frac{1}{e^{\hbar\beta\omega_n}-1} + 1.$$

$$\langle q_n(t)q_n(t')\rangle = \frac{\hbar}{2m\omega_n}\left\langle (b_n^2 e^{-\mathrm{i}\omega_n(t+t')} + b_n b_n^\dagger e^{-\mathrm{i}\omega_n(t-t')} + b_n^\dagger b_n e^{\mathrm{i}\omega_n(t-t')} + {b_n^\dagger}^2 e^{\mathrm{i}\omega_n(t+t')})\right\rangle_\beta,$$

$$\langle q_n(t)q_n(t)\rangle = \frac{\hbar}{2m\omega_n}\coth\frac{\beta\hbar\omega_n}{2}.$$

$$\begin{aligned}\langle q_n(t)q_n(t')\rangle =& \frac{\hbar}{2m\omega_n}[(\bar n+1)e^{-\mathrm{i}\omega_n(t-t')} + \bar n e^{\mathrm{i}\omega_n(t-t')}]\\ =& \frac{\hbar}{2m\omega_n}[(2\bar n+1)\cos\omega_n(t-t') - \mathrm{i}\sin\omega_n(t-t')]\\ =& \frac{\hbar}{2m\omega_n}[\coth\frac{\beta\hbar\omega_n}{2}\cos\omega_n(t-t') - \mathrm{i}\sin\omega_n(t-t')]\\ \equiv& \nu_n(t-t') + \mathrm{i}\eta_n(t-t').\end{aligned}$$